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Understanding User Perceptions and the Role of AI Image Generators in Image Creation Workflows

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Abstract

The emergence of AI Image Generators (AIGs) has transformed image creation, making it more accessible to generate customized images from simple text prompts. While HCI research has explored the applications of text-to-image generation, the role of AIGs in visual content creation workflow remains relatively underexplored. To address this, we conducted in-depth interviews with 26 end users who had experience across 14 different AIGs and investigated users' adoption and perceptions of AIGs and AI's role throughout the entire workflow. Key factors examined include user goals, initial vision, tool integration, and decision-making. Our results indicated that functional goals often drive cross-tool integration to achieve desired outcomes, while in use cases motivated by recreational goals, the usage of AIGs influences the social implications of image sharing. We concluded with four distinct use cases, each highlighting how AIGs are integrated at different stages of the creative process based on varying user goals and visions.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

Keywords

Text-to-Image, Image Creation, Creation Workflow, Generative AI

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1 Introduction

AI image generators (AIGs) have democratized the process of image creation, enabling end users, including those without expertise in model training, to generate novel and high-resolution images from simple text prompts. This has made image generation more feasible, efficient, and easy [33]. Notable AIGs like Midjourney[1], DALL-E[2], and Stable Diffusion[3] have attracted millions of users and are widely adopted for creative practices in various domains [68], such as art and design [12, 28, 38, 44]. With the increasing number of end users adopting AIGs, it is worth understanding how these tools are utilized in the creative process, especially within the entire image creation workflow.

Prior research has explored the role of general generative AI (GenAI) tools in user workflows [61]. However, unlike other types of GenAI tools, AIGs have distinct characteristics within the creation process. These differences arise from the richness of visual cues inherent in images, which can embed informational content and deliver aesthetic appeal [46], as well as the underlying motivations driving users to achieve specific visual outputs. Although previous studies [16, 51] have proposed models of the creative process in visual content creation, it remains unclear from a focused perspective on how current AIGs support these processes from conception to completion, as well as subsequent actions taken with the generated outputs, such as sharing. Therefore, a deeper exploration of the unique visual content produced by various AIGs and their role in the creation workflows is needed.

Recent studies on AIGs in the HCI field has focused on user perception [39, 40, 70, 73], interactive prompting experiences [39, 52], co-creation [19, 22, 30, 36], and AIG communities [57, 70]. While AIGs have been shown to accelerate the creative process and produce high-quality images, studies have also highlighted the discrepancies between user intentions and generated outcomes [12, 52, 70]. This misalignment presents challenges for users, particularly those without extensive experience, who may struggle to effectively control the results. Previous research has attempted to address this issue by examining prompting and interaction [39, 52], uncovering the underlying mechanisms of AI systems [49], improving prompting interfaces [17, 24, 77, 81], and expanding input conditions [42, 48, 78]. However, these studies primarily concentrated on a narrow set of AIGs, limiting the understanding of real-world user experiences. As more tools integrate T2I and new AIGs emerge, various models [10], features, output quality and styles [25] are adoptable. It is increasingly important to consider the diverse range of AIGs in various contexts of adaptability. Moreover, there is a lack of micro-level insights into how users interact with AIGs, such as their decisions, behaviors, or emotional responses throughout the processes, which could potentially deepen understanding of the

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[1] https://www.midjourney.com/home
[2] https://openai.com/index/dall-e-3/
[3] DreamStudio is the web interface for stable diffusion powered by Stability AI. https://dreamstudio.ai/

misalignment. Thus, a broad and deep investigation into the diverse usage of AIGs in creation workflows and processes is crucial.

Our research aims to extend the existing research by conducting a detailed investigation on the entire creation workflows, from creation to post creation. Specifically, we explore user goals, their initial vision of the desired outcome, decision-making in image creation, user behaviors in interaction, and post-creation activities. This research seeks to contribute to the ongoing debate on end users' perceptions of various AIGs. In doing so, we investigated the following research questions:

- RQ1: How do AI image generators influence user image creation workflows?
- RQ2: What decision-making processes do users experience in the image creation process?
- RQ3: What actions do users typically take after creating AI-generated images, and what are the underlying motivations driving these behaviors?
- RQ4: What factors influence users' decisions to adopt AI image generators, and how do they perceive the use of these tools?

We conducted in-depth semi-structured interviews with 26 amateur end users who had prior experiences across 14 different AIGs to address these questions. In the study, participants were also asked to complete a think-aloud image creation task, enabling us to capture their usage patterns and decision-making. Drawing on Botella et al.'s [16] six stages creative process, we identified four distinct use cases, which varied from user goals and initial vision: strategic, discovery, interest, and explorational-oriented. These use cases illustrate how AIGs are involved in different stages of the creative process. Our findings suggested that participants with functional goals often combine AIGs with other visual design tools by weighing trade-offs to achieve desired outcomes. Conversely, recreational goals influence the social implications of image sharing and the fulfillment of social connection throughout the creative process. In summary, our study makes the following contributions:

- We categorize usage scenarios into four distinct use cases based on their goals and initial visions: strategic, discovery, interest, and explorational-oriented. These use cases demonstrate the diverse ways in which AIGs are integrated into the creative process, including tool selection, desired outcomes, and post-creation actions.
- We identify three key decision-making aspects within the generalized user pattern: vision development and prompting, outcome evaluation, and adjustment and fine-tuning.
- We explore diverse post-creation actions, with a particular focus on sharing. We recognize social implications of sharing generated images, including underlying motivations for effective communication, fostering connection, and utilizing them as conversation starters.
- We identify shared perceived benefits and challenges across 14 AIGs and contain some individual differences.

2 Background and Related Work

We begin by providing background information on text-to-image generation (T2I) and AIGs to establish the study focus. Given our concentration on the creation workflow, we review prior literature on creative processes in visual content creation and creativity support tools. Subsequently, we focus on AIGs, understanding user perceptions, the strengths and limitations in supporting image creation, and user interactions with these tools. Finally, we explore the later stages of the workflow, specifically reviewing the application and sharing of AI-generated images.

2.1 Text-to-Image Generation and AI Image Generators: Background and Mechanism

AI image generators, also known for text-to-image generation, are advanced products built on deep learning models that enable users to create high-quality images from text prompts or other input modalities. Prominent platforms include Adobe Firefly [2], Craiyon [3], DALL-E 2 [4] [4], DALL-E 3/ChatGPT [5], DreamStudio [6], Midjourney [7], Runway [8], and others. While underlying technical details may vary, their core function remains T2I generation. Some AIGs also offer additional features like sketch-to-image or image-to-image [54, 83] for greater control and detailed customization. Beyond input methods, variations in user interface interaction design, user experience, accessibility, performance, and image quality have been observed across platforms [10, 25].

The success of T2I generation has been driven by the pre-trained multimodal encoder, CLIP (Contrastive Language-Image Pretraining), which connects textual and visual concepts by training on large-scale natural language and image pairs [63]. Besides, the recent diffusion-based image generation methods [37, 65, 67, 69] have further boosted neural networks' capability to generate unique and diverse images. While studies have shown novel and high-quality image generation results, a common challenge still remains the misalignment between input conditions and generated images. The technical field has made efforts to tackle this issue. For instance, the DALL-E-3 model [11] improves prompt-following ability through enhanced image captioner for training the generator's dataset. Researchers in the HCI field have also focused on addressing this challenge, particularly in attempting to resolve the complexities of prompt engineering from human perspectives [17, 24, 62, 77, 78].

By examining AIG mechanisms and features, we have gained a deeper understanding of the unique nature of AI-generated images. Reduced accessibility barriers have empowered end users to create synthesized images and accelerate the creative process without knowing the technical underpinning. The generalized usability of AIGs has also fostered increasing creative applications.

2.2 Visual Content Creation and Creativity Support Tools

Creativity and creative processes have been extensively discussed by scholars. Rhodes's [66] 4P model — person, process, press, and products — provided a broader view of human creativity and highlighted the interaction between individual characteristics, ideas on creative works, and environmental factors. In the context of visual art, research has further explored the factors that influence creativity. Studies concluded the impact of cognitive, conative, emotional, and environmental factors on creative practices [15].

[4] DALL-E 2 is no longer in service as of April 2024.

The iterative or dialectical nature of the creative process has been recognized [13], and various models have been proposed to describe it [50]. With a focus on visual content, Mace and Ward [51] identified four stages in art: conception, idea development, making, and finishing. This process moves from vague concepts to tangible entities, requiring continuous decision-making and evaluation. Similarly, Botella et al. [16] propose a six-stage model, including general idea and vision, documentation/reflection, first sketches, testing forms/ideas, provisional objects/drafts, and final work/series. Scholars have highlighted that the creative process involves a series of thoughts and actions, which is not fully sequential or linear but can be iterative [16, 50]. By delving into the creative process of visual content, particularly Botella et al.'s [16] six-stage model covering detailed decision-making from the initial vision to the final outcome. The model offers a foundation for exploring the role of supporting tools at each stage.

Creativity support tools can positively influence users during the creative process [31]. Computational tools can enrich human creativity, particularly in discovery and innovation [72], as well as aid designers in self-expression and reflection, fostering both divergent and convergent processes [14]. While the impact of tools on creativity is evident, individual perceptions of their value may be different. Palani et al. [60] investigated the adoption of creativity support tools across fields and revealed that practitioners prioritize functionality, integration, performance, and user experience in their decision-making. Additionally, their study also emphasized the importance of tool compatibility within users' existing workflows. Understanding these perspectives allows for deeper investigation into user perceptions and the role of creativity support tools within the workflow.

2.3 Generative AI in Image Creation

AI-powered creativity support tools are increasingly integrated into design and creative processes, often assisting in the later stages like develop and deliver [43]. In a broader GenAI context (i.e., text, image, code, etc.), Palani and Ramos [61] examined the role of these tools in workflows, finding that practitioners value agency and carefully consider trade-offs from different aspects. However, the specific role of AIGs in workflows remains unclear due to their distinct visual characteristics. Prior research [57] has outlined the creativity landscape of T2I generation and explored dimensions including practitioners (i.e., person), prompt engineering and image curation (i.e., process), T2I ecosystem (i.e., press), and digital images (i.e., product) based on Rhodes's 4P model [66]. We delve into these dimensions as outlined below:

2.3.1 User Perceptions. Previous studies have investigated the perspectives of various populations on T2I adoption, including visual artists [40], game industry professionals [73], students [30], and users from diverse backgrounds in online communities [39, 70]. With a specific center on DALL-E-2 and Midjourney, key factors motivating users to adopt T2I are curiosity, entertainment, work-related purpose, self-expression, and artifact design [30, 70]. Research indicates that T2I is beneficial for ideation [18, 19, 64, 82], expanding imagination through variation [12, 38, 53, 59], reflection [38], and inspiration [19, 82]. It is particularly valuable in the early stages of the creative process [44] and accelerates design iteration [40, 53]. Although the strengths are recognized, several perceived challenges and concerns also arise. The primary challenge lies in the difficulty in prompting and the misalignment between intention and outputs [12, 52, 70]. Translating individuals' conceptual textual descriptions into information interpretable by the AI model remains a complex task. Additionally, limited user controls within the creative process [45], divergent interpretations of generated images [27], and ethical concerns [35] are key areas of gaining consideration.

2.3.2 Creation Process and Interaction. In the image creation process, prompting plays a critical role in T2I generation. It can also be considered an artistic process rather than merely inputting textual descriptions [20]. Several studies have concentrated on user prompting structure. For instance, Sanchez [70] found that prompts often include subject, medium, influence, light, color, composition, detail, and context. Oppenlaender [58] identified similar elements but focused on modification. For specific AIGs like Midjourney, other research [52] has identified five common prompt structures, incorporating descriptive sentences, templates, overview and detail, chunks, and word sequences.

To understand the interaction between users and AIGs at a micro-level, Mahdavi Goloujeh et al. [52] investigated the prompt journey in Midjourney, revealing an iterative cycle of prompting, evaluation, and refinement. Their study also highlighted the specific evaluation criteria and refinement strategies in the creation process. A similar process was observed in DALL-E 2 usage, where Kim et al. [39] noted that users gradually adapted to the AIG through continued interaction, refining their prompts to achieve optimal results. While the processes are important, a UX research report [55] emphasized that user goals significantly shape the journey. They identified two primary goals in Midjourney usage, inspiration and deliverable, which influence distinct creative processes and AIG involvement. While these studies provide insights into the creation process, they mainly focus on specific AIGs. Additionally, a deeper understanding of user behavioral aspects of interaction across various AIGs remains limited. Further investigation is needed to investigate how different AIGs contribute to the process and entire workflow, ranging from user goals to post-creation actions.

2.3.3 Manipulation and the Sharing of AI-Generated Images. AI-generated images have diverse applications, including storytelling, communication, visualization, etc. For instance, they effectively convey emotions for surreal scenarios like dreams [76], bridge communication gaps in collaborative settings [23], and facilitate real-time visual idea exchange [40, 64]. Additionally, they were used for visualization tasks, such as describing illnesses or representing specific visual scenarios [75]. While the contextual usage of AI-generated images has been explored, there's little known about users' underlying motivations for using or sharing them. Photographs, as a storytelling medium, shape personal lives and relationships [9]. Prior studies have shown that individuals engage and share images for various reasons. Frohlich et al. [32] found that photographs act as narrative catalysts, prompting conversations, evoking emotions, and providing enjoyment. Photo sharing facilitates co-present experiences, enabling individuals to relive memories. Other research emphasized the role of digital tools in augmenting the social motivations for image sharing, maintaining

connections, and enhancing the sense of social presence [26]. Voida and Mynatt [74] identified six themes of sharing via instant messaging, such as amplification, narrative, awareness, local expression, invitation, and object/instrument, highlighting the complex role of images in computer-mediated communication. Existing literature provides insights into image sharing motivations and their benefits for social interactions. However, given the unique synthesis and highly customized characteristics of AI-generated images, further research into post-creation actions and sharing generated content is worth exploring. Building on prior work, our study aims to explore the role of various AIGs in the entire creation workflow, focusing on goals, creative process, and post-creation activities, and user overall perspectives. Analyzing the process from its initial stages to its finals provides a comprehensive understanding of AIGs and their distinct characteristic in image creation.

3 Method

To address our research questions, we conducted in-depth semi-structured interviews and a think-aloud image creation task with 26 amateur end users. Unlike professional users, these amateur participants may vary in their motivations (e.g., not finding financial gain), skill sets [29], and prior experience utilizing text-to-image generation in AI image generators. We aimed to elucidate their distinct goals, the impact of AIGs on their image creation workflow, and their overall user perceptions.

3.1 Participants

We used both offline and online approaches to recruit eligible interviewees. Recruitment flyers were distributed at a large university in the USA, as well as its surrounding city. As for online strategies, recruitment information was disseminated through the university's participant recruiting system and university-related online communities — Discord, GroupMe, and Slack — to reach a wider audience. All participants were required to 1) have experience using existing AIGs, specifically T2I generation; 2) have no professional expertise in graphic design or art with AI and do not create visual content as their primary job; and 3) be aged 18 and above. Prior experience with image-to-image (I2I) features was considered a plus.

A total of 26 participants were recruited, all students from a variety of backgrounds and majors. These participants represented a diverse range of demographic characteristics, including gender (19 females, 7 males), age (ranging from 18 - 39 with one preferring not to specify; M = 23.16; SD = 4.88) (see Table 1), education level (10 graduate, 16 undergraduate), and native languages (see Table A.1). Additionally, we inquired about participants' experience with AIGs as well as the visual design tools they typically use (see Table 1). They reported experience using 14 different AIGs, with Dall-E-2 (n = 16), Midjourney (n = 12), and Canva AI (n = 7) being the most frequently mentioned generators in this study. Seven participants also reported experience with I2I generation features within existing AIGs such as Midjourney, Fotor, Dall-E-3, and Canva AI. All participants demonstrated a desire and interest in creating digital images using their familiar visual design tools.

3.2 Study Procedure and Interview Protocol

We conducted in-person semi-structured interviews with participants in a laboratory space from September to December 2023. The studies were conducted in English and lasted between 30 and 82 minutes, with an average of 52 minutes. Prior to the interviews, all participants provided informed consent and agreed to audio recording. To gather background information, we asked participants to complete pre- and post-interview surveys using Qualtrics. These surveys collected demographic data, exploring participants' experiences with AIGs, visual design tools, and their understanding of Artificial Intelligence. The interview protocol focused on understanding participants' perspectives on AIGs and their decision-making processes when using these tools to produce images. Upon completing approximately one hour study, we offered $15 USD as compensation for each participant. The study was approved by the university's Institutional Review Board (IRB). Our study contained three following sections, incorporating an interview and an image creation task:

3.2.1 User Experience with AI Image Generators and Tool Selection. This section explored participants' general experiences, motivations, goals, and the process of using AIGs, with a particular focus on T2I generation. Firstly, we asked about their recent image creation experiences **without** using an AIG, including image type, tools, and the tool selection criteria in different contexts. After explaining the definition of AIGs, we delved into their experiences **with** these generators, focusing on similar questions about their recent image creation experiences and the factors influencing the generator choice. To understand tool evaluation in image creation, we asked questions such as "What's the difference between using AI image generators and other graphic editors without the support of generative AI?" Furthermore, we investigated the factors that influence AIG adoption, including user goals, frequency of use, types of images, and post-creation activities. For those who shared their generated works, we further explored content, methods, recipients, and sentiments associated with the sharing experiences.

3.2.2 Think-Aloud: Image Creation Task. In order to understand participants' decision-making processes during their usual image creation with AIGs, we employed the think-aloud method [21], closely observing participants' interactions with AIGs, taking notes, and asking follow-up questions throughout the entire task. Participants were asked to verbalize their thoughts and describe their processes while creating generated images. This allowed us to capture their behavioral patterns and perceptions across various creative stages, incorporating their motivations and goals, vision of the desired outcome, fine-tuning strategies, satisfaction, and intended actions after the creation. The image creation task required participants to follow the following steps: 1) select a preferred AIG based on familiarity or most frequent use; 2) develop an initial vision for the desired image; 3) sketch their envisioned vision on an iPad to visually represent their ideas and then verbally describe their imagination. We included this sketching activity to explore how grounding their initial vision might influence their approach to this task, even though it may not fully reflect their usual process when using AIG; 4) go through their usual creation process, generating images using their chosen generator and any additional tools they

Table 1: Participants' demographic information, along with the usage of AI image generators and visual design tools.

ID	Gender	Age	Adoption[a]	Avg Usage Frequency[b]	AI Image Generators[c]	I2I Usage[d]	Top Visual Design Tools[e]
P01	M	39	H1 2023	Twice or thrice a month	MJ, DE2, SD	×	Adobe Photoshop*
P02	F	×	H1 2023	Twice or thrice a month	MJ, DE2	○	Adobe Photoshop*
P03	F	30	H1 2023	A few times	DE2	×	Microsoft PowerPoint*
P04	M	27	H2 2023	A few times	SD	×	Keynote*
P05	F	19	H2 2022	Once a month	CAI, MJ, BI	×	Canva
P06	F	23	H1 2023	A few times	MJ, DE2, CAI	×	Canva
P07	F	23	H2 2023	Once a week†	RW, DE2, MJ	○	Adobe Photoshop*, Procreate*
P08	F	21	H1 2023	Twice or thrice a month	DE2, CR, DS	×	Canva*
P09	F	19	H1 2023	A few times	MJ, DE2, SD, BI	×	Canva*, Art set (APP)*
P10	F	18	H1 2023	Once a month	CR, CAI, DI	×	Canva
P11	M	20	H2 2022	A few times	DE2	×	Photopea*
P12	F	19	H1 2023	A few times	DE2, FO	○	Canva*
P13	F	29	H2 2022	Once a week	MJ, CAI, DE2, BI, AF, FO	○	Canva
P14	M	25	H2 2022	Twice or thrice a month	DE3, MJ, SD, AF	○	Figma*
P15	F	29	H2 2023	A few times	AF, BI	×	Microsoft PowerPoint*
P16	M	20	H2 2023	A few times	DE2	×	Microsoft PowerPoint*
P17	F	29	H2 2022	A few times	DE2, CAI, MJ	×	Keynote*
P18	F	21	H2 2022	Twice or thrice a month	TT, DI	×	TikTok
P19	M	20	H2 2023	Once a week†	BI, DE3, SD, CR, MJ	×	Adobe Illustrator*
P20	F	21	H2 2022	More than once a week	DE3, MJ, DE2, CR, CAI	×	PixilArt*
P21	F	21	H2 2023	More than once a week†	DE3	○	Microsoft PowerPoint*
P22	M	22	H2 2023	A few times	GC	×	Draw.io*
P23	F	21	H2 2022	A few times	MJ	×	Microsoft PowerPoint*
P24	F	20	H1 2023	A few times	DE2	×	Canva*
P25	F	20	H2 2023	Twice or thrice a month†	DE3, DE2	×	Microsoft PowerPoint*
P26	F	23	H2 2022	A few times	DE2, CAI	○	Canva

[a] Adoption refers to when participants began using AIGs. H1: the first half of the year, H2: second half. [b] Avg usage frequency indicates the average use of AIGs over the past 6 months. [c] AI image generators refer to the generators that participants have ever used. The first mentioned generators are those that participants have used most frequently. Adobe Firefly (AF), Microsoft Bing (BI), Canva AI Image Generator (CAI), Craiyon (CR), Dall-E-2 (DE2), Dall-E-3/ChatGPT (DE3), Deep AI (DI), DreamStudio (DS), Fotor (FO), Gencraft (GC), Midjourney (MJ), Runway (RW), Stable Diffusion (Online) (SD), AI Greenscreen on TikTok (TT). [d] I2I Usage: Image-to-image usage. [e] Top visual design tools indicate participants' most frequently used tools. † The data may not accurately reflect average usage frequency over the past 6 months due to their adoption time. * Participants have not used the built-in image generation features of the tool at the time the study was conducted.

typically apply; and 5) express their feelings about the generated images and subsequent intended actions. For those planning to share their images, we followed up by asking sharing method, intended recipient(s), etc.

3.2.3 The Overall Perception of AI Image Generators. This concluding section involved exploring participants' perceptions of AIGs, including perceived benefits, challenges, and understanding of the underlying image generative models. We delved into how end users view and utilize AIGs in a broader context.

3.3 Data Analysis

Audio recordings were first transcribed. We then employed an inductive approach to analyze the transcripts. The first author began by applying open coding [71, 79] to all data using ATLAS.ti 24.1.1, and then organized the codes into emerging themes. Through regular discussions and iteratively revisiting the transcripts, we refined our understanding of the codes and themes. During the analysis, we also identified 50 usage scenarios based on 26 participants' self-reported prior experiences with AIGs, with some sharing more than one example. To better capture the sequence and context of AIG usage, these scenarios were categorized and analyzed using a process-oriented approach. Further analysis of the usage scenarios drew on Botella et al.'s [16] six stages of the creative process and was organized into use cases. For the think-aloud image creation task, we incorporated multiple data types and concentrated on the process during the analysis, see Section 3.3.1. Finally, after carefully reviewing all the themes, subthemes, and corresponding quotes again, we reached a consensus on the refined findings and provided key quotes to support the presented results.

3.3.1 Image Creation Task Analysis. To analyze the decision-making process and the micro-level interactions with AIGs throughout the think-aloud task, we examined their 1) sketches, 2) transcripts, 3)

chosen AIGs, 4) prompts, 5) generated images, and 6) researcher notes. The first author conducted an in-depth review and organized multiple modalities of data according to each participant's process (see Figure B.1). Following the process-oriented approach, we sought to understand the general usage patterns shared across participants. The first author extracted key information, such as sketches, prompts, and generated outcomes by following the mechanism of AIGs to visually represent their process in diagrams. We used different colors to represent various data types and arrows to indicate the creation flow (see Figure B.1). We collaborated with two external coders who exclusively focused on analyzing the image creation task and diagrams. For the initial round of categorization, the first author and the two external coders independently coded the data and grouped participants into patterns. Subsequently, we iteratively revisited data to develop a coherent interpretation of the diagrams. Finally, we collaboratively reviewed all data and exchanged perspectives to reach an agreement on the identified user patterns and relevant findings.

It is important to note that we excluded three participants from the image creation analysis due to differing user goals and processes. Specifically, two participants used T2I generation (i.e., RW and TT) as part of their process but aimed to create videos as their final outcomes, while another participant relied on ChatGPT for developing the initial vision. In summary, our participants utilized 9 different image generators, including AF (n=1), BI (n=1), CAI (n=1), CR (n=2), DE2 (n=7), DE3 (n=3), GC (n=1), MJ (n=6), and SD (n=1) for the image creation task.

4 Results

Our results provide a comprehensive analysis of the use of AI image generators, covering both high-level and detailed aspects of creation workflows and user perceptions. This incorporates AI involvement at different stages of the creation workflows (Section 4.1), a closer examination of a generalized usage pattern and decision-making in creation process (Section 4.2), post-creation actions, particularly social implications of sharing generated images (Section 4.3), as well as general user perceptions across multiple platforms (Section 4.4).

4.1 Four Use Cases of AI Image Generators in Creation Workflows

To obtain an overview understanding of the role of AIGs and their influence on user image creation workflows (RQ1), we analyzed 50 self-reported usage scenarios from participants' prior experiences and categorized them into four distinct use cases: 1) strategic-oriented, 2) discovery-oriented, 3) interest-oriented, and 4) explorational-oriented (see Section 3.3). Each use case varies in terms of goals (i.e., functional or recreational), initial vision clarity (i.e., clear or abstract), and the process of achieving the desired outcomes. Additionally, we considered tool usage, integration, and post-creation actions to understand how AIGs support these processes. Building on Botella et al.'s [16] model of the six stages creative process, we followed their descriptions and slightly modified their naming to Initial Vision, Vision Reflection, First Attempt, Idea Testing, Refining Prototypes, and Final Work. By mapping each use case to these stages, we gained a deeper understanding of how AIGs can be involved in different stages of creation workflow, see Figure 1. Note that this analysis is based on usage scenarios, rather than individual differences.

4.1.1 Strategic-Oriented. In the strategic-oriented use case (n=11), participants have a clear vision of the desired outcomes and strategically prioritize creating high-quality and usable visual content for functional purposes. AIGs play a central role in the mid-stages process. After fine-tuning or retouching with familiar tools, they often share or integrate the final outcomes into their existing workflows.

Specifically, participants are driven by practical motivations, such as preparing visuals for a talk or creating visual references for slides, with the goal of creating polished visual content. With a clear and concrete vision, participants actively used AIGs to fine-tune images, which heavily involved AIG in the First Attempt and Idea Testing stages. *"I was trying to develop icons based on some prompts, and these items basically did a demonstration of the farming process. [...] I had a particular image in mind when I wanted that kind of graphic, so I used a combination of Bing, Canva, and Midjourney." (P13; MJ, CAI, DE2, etc.)* Additionally, participants often make minor edits to generated images using familiar visual design tools. These final images are then integrated into their existing workflows. *"When I wanted to write my article on the internet, I tried to describe a very detailed image of two people, one standing on the left and one on the right. [...] After it generated the background, I used Photoshop to add words to it because the quality of the drawings was pretty good, much better than my hand drawing." (P01; MJ, DE2, and SD)* Additionally, the final outcomes are typically shared with others or used directly, reflecting participants' intention to produce content ready for practical applications.

4.1.2 Discovery-Oriented. In the discovery-oriented use case (n=18), participants start with abstract initial visions and discover visuals that can practically support their functional purposes. AIGs play an important role in the early stages, facilitating ideation, exploring possibilities, and visualizing concepts. The generated outcomes often serve as inspiration that leads to integration into existing workflows or further retouching using familiar tools.

To elaborate, unlike the strategic-oriented case, discovery-oriented begins with an undefined vision and focuses on concretizing ideas and exploring alternatives while still aiming for practical final outcomes yet not limiting to AI-generated content. For instance, P06 aimed to complete a poster in the end and used AIGs in her workflow. *"I tried to give some prompts, but what I gave was to create a poster, poster name, and topic. I hope that AI can help me design the topic based on what I have said, but obviously, my prompt was not clear enough." (P06; MJ, DE2, and CAI)* Furthermore, participants actively used AIGs to develop ideas and refine concepts, which heavily involved AI in the Vision Reflection and First Attempt creative stages (see Figure 1). They focus on generating ideas and visualizing concepts rather than extensive interaction with the AI tools. *"One of our projects was a game. It was a platformer sort of like Mario. Since it's a game, you have to draw things on the screen. You need to have a player character or sprite background. [...] AI image generation was a way to add (graphics)... I wouldn't say they're particularly impressive graphics, but they're good enough for us to ship something out." (P11; DE2)* These inspirational generated outcomes prompt participants to integrate the visuals into their existing workflows rather than refining them further in the AIGs. *"That's not even the*

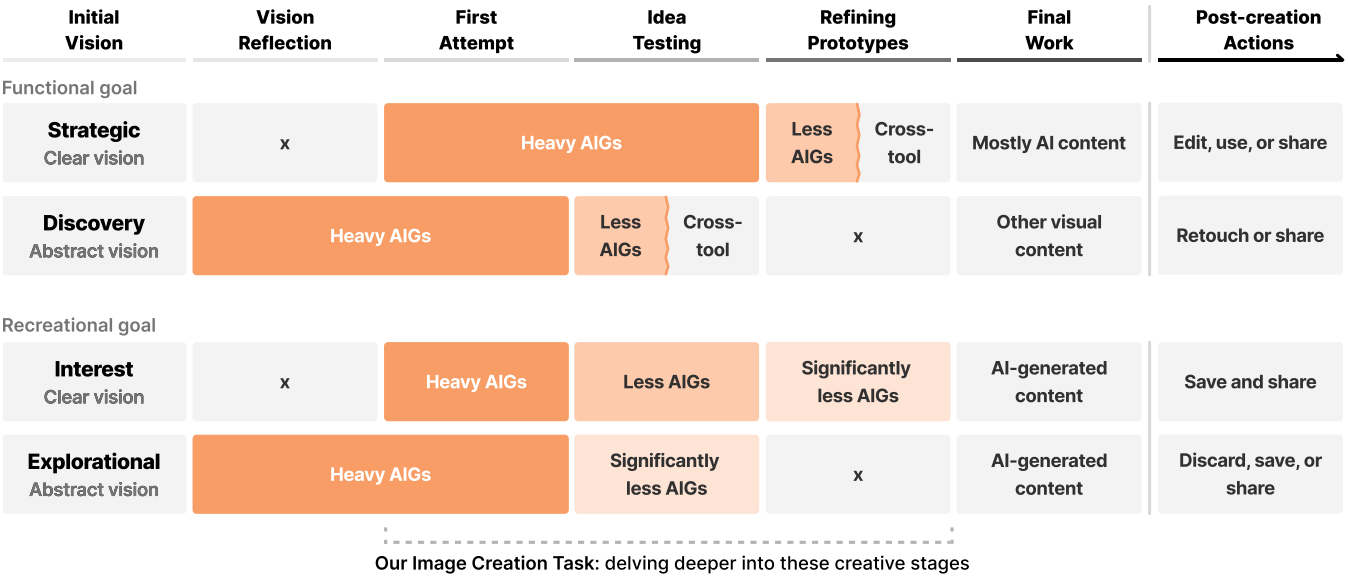

Figure 1: Four use cases — strategic-oriented, discovery-oriented, interest-oriented, and explorational-oriented — illustrate how AI image generators are involved at different stages of the creation workflow. The workflow includes both six stages of creative process and the subsequent post-creation actions. The depth of the orange color indicates the level of AIG involvement.

final product. That image just reminded me of something else, and I wanted to add another element to the picture, so I added." (P08; DE2, CR, and DS) This motivation frequently leads to retouching or further creation using other visual design tools. *"I used it as inspiration to try to create my own design, so I would try to mimic some aspects of it (generated image) with my own design." (P10; CR, CAI, and DI)*

4.1.3 Interest-Oriented. In the interest-oriented use case (n=10), participants have a clear vision and are driven by recreational motivations, like personal interest, to create potential images for sharing. They focus on producing polished final work with AIGs playing a role throughout the process and some fine-tuning.

Particularly, participants' motivations are driven by passion or personal interest in creating certain types of images, such as emoji, meme, or wallpaper. *"Me and my partner were trying to make an emoji, a new Apple design emoji that had big duck lips and face, because there's no emoji that has the comical feature like that, so we were playing with DALLE together." (P25; DE3 and DE2)* In this case, most participants have concrete visions and rely on AIGs as the primary tool throughout the entire process, from First Attempt to Final Work stages. *"I typed in a prompt where it was a mix between Barbie and Fairy movie dress. [...] It turned out really nice, and it was kind of a dream dress design. I have an image of it now, so maybe one day it will come to fruition somewhere." (P05; CAI, MJ, and BI)* Moreover, they invest relatively less effort in fine-tuning or retouching images compared to strategic-oriented case, often considering the generated outcomes as final products. *"I remembered the summer with one that I liked particularly, I printed and stuck it on my wall." (P08; DE2, CR, and DS)* Users even seek visual content that is ready for sharing: *"I shared a lot of them (generated images) with my friends or family, just to share the cozy feeling." (P20; DE3, MJ, DE2, etc.)*

4.1.4 Explorational-Oriented. In the explorational-oriented use case (n=11), participants are driven by recreational purposes like curiosity, with a desire to explore AI's capabilities through abstract initial visions. They focus on the exploration process rather than prioritizing practical outcomes, with AIGs playing the main role in the early stages. Although most generated images are discarded, unexpectedly interesting results may lead to sharing.

This use case is primarily driven by curiosity about the capabilities of AIGs. Participants in this use case approach with abstract vision and are not necessarily looking for practical or usable outcomes. Instead, they are interested in experimenting and interacting with AI. *"I put in music artists that I listen to, so I wanted to see what AI would produce." (P10; CR, CAI, and DI)* AIGs also play a role in the Vision Reflection and First Attempt stages, facilitating the visualization of concepts. *"One of the trends (on TikTok) actually involved [University name], putting your college name into the AI. It generated an image based on the campus, and they're kind of realistic but in an abstract way." (P18; TT and DI)* However, as participants prioritize the process, we identified that the generated images are often discarded. *"I just generate and see: this is good, this is bad, this is useful. I think I'll end up there." (P02; MJ and DE2)* Unless generators produce unexpectedly interesting results, there may be the possibility of sharing AI-generated images with others. For instance, P18 further shared the experience on TikTok, *"It was an interesting experience for it (TikTok) to generate. It amused and fascinated me how it could generate this image. [...] I made the video to post it, to see how it generated the image." (P18; TT and DI)*

4.2 Decision-Making in Image Creation Process

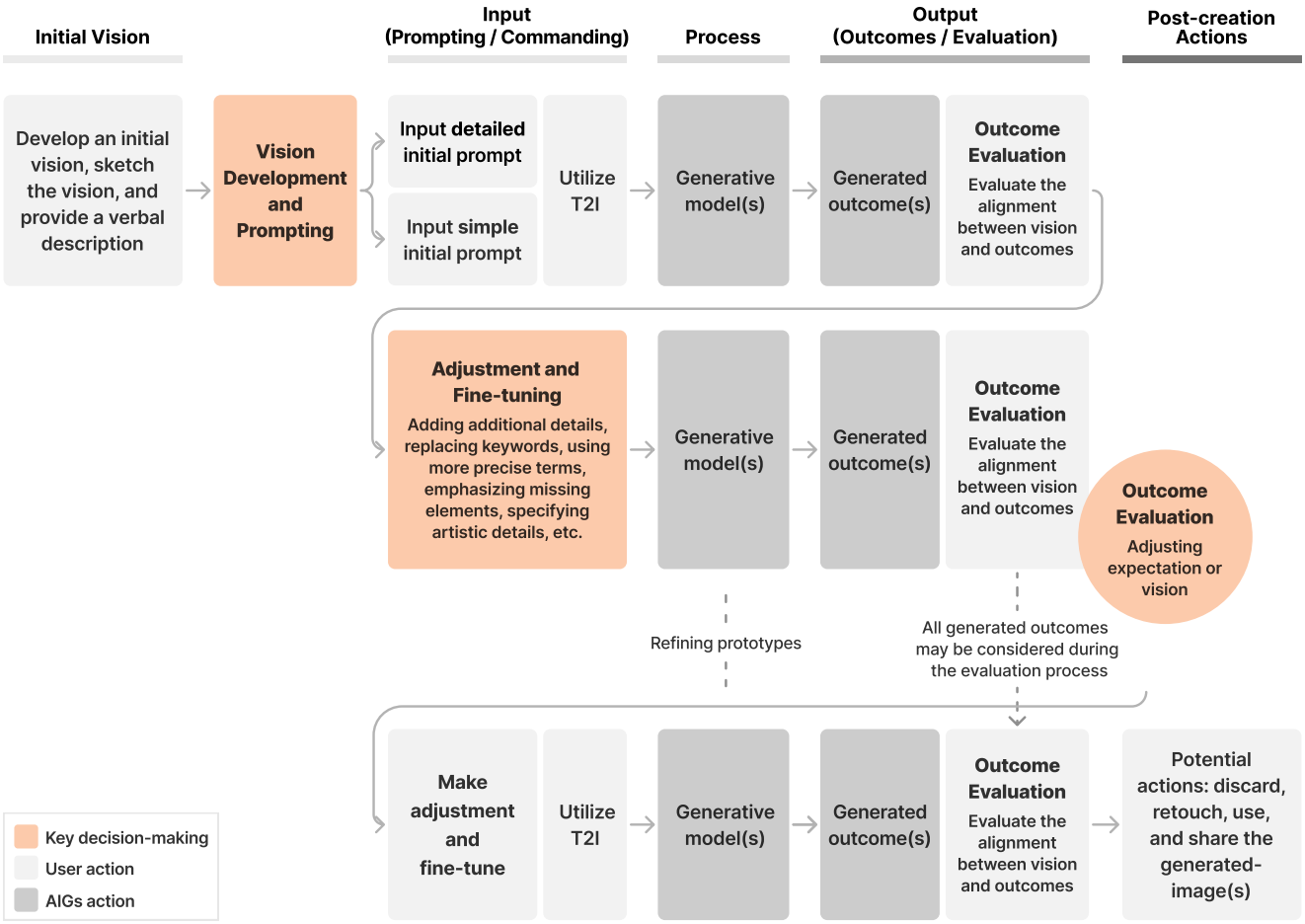

Figure 2: A generalized usage pattern in the image creation process emerged across 9 AI image generators by covering initial vision, input, generative model processing, output, and post-creation actions. The usage pattern reveals three key decision-making aspects in image production: 1) vision development and prompting, 2) outcome evaluation, and 3) adjustment and fine-tuning. These aspects highlight the individual differences in cognitive and behavioral patterns during the creation process.

To delve deeper into the impact of AIGs, we aim to understand the decision-making in image creation process (RQ2). We seek to conduct an in-depth analysis of user behaviors and interactions with these tools, particularly from the "First Attempt" to "Refining Prototypes" stages (see Figure 1). Through our observations and analysis of think-aloud image creation task, we identified **a generalized usage pattern** shared across 9 different generators. This pattern spans from initial vision to post-creation actions, as illustrated in Figure 2. Our analysis revealed **three key decision-making aspects**: 1) vision development and prompting, 2) outcome evaluation, and 3) adjustment and fine-tuning. This usage pattern reflects an iterative process of aligning users' vision with generated outcomes, leading to temporary final results and subsequent actions.

4.2.1 Vision Development and Prompting. As described in Section 3.2.2, since we had asked our participants to articulate an initial vision of the desired image, twenty-three participants had a relatively clear vision of the desired image. To translate their ideas into actual text input, eighteen participants mentioned that they tended to start with getting something simple, focusing on key elements or subjects in their initial prompt. *"I normally start with get something out and see what I can adjust." (P04; SD)* Furthermore, participants developed personalized strategies for crafting prompts based on their prior experiences. *"The prompt works best when it is not complicated and just with several short terms, keywords, key adjectives, and nouns. I'm not trying to describe it in detail because that doesn't always work well." (P01; MJ, DE2, and SD)* All participants relied on T2I as their primary or sole input strategy, with only three utilizing predefined image styles (i.e., commanding) offered by AIGs, such as Adobe Firefly, Gencraft, and Craiyon.

However, we observed a discrepancy between the initial vision and the initial prompt for some participants. These initial

inputs could generally be categorized as either **detailed or simple prompts that influence subsequent decision-making processes.** Fifteen provided detailed initial prompts that covered various aspects such as subjects, actions, scenes, and artistic details, but eight offered simpler ones that did not fully capture their desired vision as described. For instance, P24 was looking for: *"this is an organization that provides financial support to athletes which is called [Name]. Essentially, what the logo shows us is the name and an image of someone passing a baton to another person, which is basically a symbol of providing support and collaboration in the field of sports." (P24; DE2)* Nevertheless, P24 only searched up "sports nonprofit" as the initial prompt. Such misalignment may require participants to make additional efforts for further evaluation and adjustment.

4.2.2 Outcome Evaluation. After AIGs processed the inputs, participants reached a stage where they assessed the generated outcome(s) based on their own criteria. Many also frequently expressed immediate preferences or dislikes for generated images. *"I look at the result and then see. Sometimes, it's not creating the image that I imagined exactly. Knowing what this AI can and cannot do in the process and trying to adjust my expectations. It's a trial-and-error thing." (P01; MJ, DE2, and SD)* However, amateur participants often lacked clear direction on how to effectively refine their further inputs. In turn, participants evaluated generated images based on their alignment with the initial vision and by **comparing different outputs within and across iterations, following a not entirely linear process.** For instance, P02 evaluated images within the same iteration by using Midjourney, *"I don't like these two because that's too complex. That's not what I want. I like these two better. It feels more realistic, but I don't like this shape so I actually prefer this one,"* as well as evaluating images across multiple iterations *"I will do it again to see if it gives me something better than these two. If not, I will go back and say that I prefer this one." (P02; MJ and DE2)* However, not all nine AIGs had corresponding interaction designs and capacity to support participants in comparing cross iterations.

In addition, we observed that when the generated outcomes did not meet their expectations, nine participants tended to adjust their expectations or revise their vision, typically after the second iteration (see Figure 2). *"Based on the prompts so far, this would be another case where I would give up and try not to use Dall-E for image generation here." (P11; DE2)* The frustration also echoed from another participant, *"I would want to change it so I could make it something more similar to my drawing." (P10; CR, CAI, and DI)* On the other hand, if the generated outcomes included unexpected elements, these could inspire participants to rethink or alter their vision. *"I'm going to give up with the multi-level thing because it's not working very well. [...] After they give me an image, sometimes it inspires me to add on, which is where the library came from." (P20; DE3, MJ, DE2, etc.)*

4.2.3 Adjustment and Fine-Tuning. After evaluating the initial outcomes, all participants expressed an intention to fine-tune the images to better align with their original vision. During this process, twenty-two participants concentrated on **making adjustments by adding further details, replacing keywords, using more precise terms, emphasizing missing elements, or specifying artistic features.** For instance, P05 specified the image layout in the fine-tuning process, *"I am going to specify that a tall waterfall flows through above the middle of it, and specify water falling to the center of the page." (P05; CAI, MJ, and BI)* Participants even changed different image styles to reach their desired vision, finding it more convenient when AIGs provided pre-categorized style option buttons. *"I'm going to change the style of the art. I want him (the subject) to be more like a candid kind of thing." (P22; GC)* Specifically, we identified four participants driven by functional goals, especially those using simpler initial prompts, who were more adaptable in adjusting their ideas or approaches to achieve desired outcomes. They experimented with different themes throughout the task. As P11 noted, *"Maybe instead of camels, I'm going to replace it with another functional programming idea and see if it does a little better." (P11; DE2)* Similarly, these four participants explored variations, like tree-like structures, by **branching out from their initial vision**. *"What I'm going to do now is try a different approach to prompt as a whole." (P24; DE2)*

4.3 Post-Creation Action: Social Implications of Sharing AI-generated Images

To investigate users' typical actions after creating AI-generated images and their motivations (RQ3), we observed various post-creation actions among use cases, including discarding, using, retouching, or sharing AI-generated outcomes (see Section 4.1). Among these, sharing stood out as particularly worth exploring actions. Our analysis revealed distinct underlying motivations with three main sub-themes that emerged in the social implications of sharing AI-generated images: 1) using temporary visuals for collaborative communication, 2) sharing interests to foster social connections, and 3) utilizing generated images as conversation starters.

4.3.1 Using Temporary Visuals for Collaborative Communication. Mostly in functional goal use cases, seven participants expressed a desire to share their ideas with others, especially in collaborative settings. Given AIGs' advantages, AI-generated images can serve as a useful medium for effective communication. For instance, P04 mentioned using visual content as an indicator to represent scenarios in meetings. Even before the demonstration, P04 recognized that the generated images would be a temporary visual aid. *"Before I shared it, I knew that it most likely would be a placeholder." (P04; SD)* Specifically, six participants used AI-generated images as temporary visuals to seek feedback or clarify envisioned visual outcomes with collaborators. *"I sent a copy to my mentor and asked her what she thought if she wanted any specific changes or edits." (P05; CAI, MJ, and BI)* Similarly, another participant was motivated to use AIGs for visual communication to enhance mutual alignment and facilitate further collaboration. *"I wanted to make sure that we (my manager and I) were on the same page. Initially, he said you could play around with it, so that's what I did. Then, I sent it back to him to ask if this is something that you envision, like the website's look?" (P09; MJ, DE2, SD, etc.)*

4.3.2 Sharing Interests to Foster Social Connections. With recreational goals, ten participants expressed a desire to create AI-generated content aligned with their personal interests and were inclined to share them with like-minded individuals. *"We were all in a group call together. Every time we made something that we thought was nice, we would kind of put it in a group chat. It was sharing our different*

tastes and art, I suppose." (P22; GC) Similarly, P01 shared generated images with friends who had similar experiences in image generation. *"For recreational use, like the ideal type, I just leave it there. When I'm bored, I just go through all my (ideal type of person) and swipe through them. Looking at them makes me feel good. Sometimes I share it with some of my friends who also like to make their ideal type." (P01; MJ, DE2, and SD)* On the other hand, participants created customized images to evoke specific reactions from others, such as P26 making an e-invite for their home party: *"we just wanted to make fun and something that's not predictable. When people see the invite, it can remind them it's fresh and funny." (P26; DE2 and CAI)* Furthermore, five participants tailored their creations to appeal to specific individuals and demonstrated a desire to connect with others through sharing. *"I usually just share something that would cater to their (my friends') tastes. I have a friend who likes cats. If I make something with cats, they'd be, 'Oh my God, look at it,' it's a cute cat image." (P20; DE3, MJ, DE2, etc.)* Through these sharing practices, we identified a tendency for many participants to share generated images with close ones, such as family members, partners, or friends, in private personal or group chats.

4.3.3 Utilizing Generated Images as Conversation Starters. For recreational purposes, seven participants indicated that they shared generated images with others, and these images intentionally or unintentionally became conversation starters. For instance, P05 shared a customized image featuring her brother's name to attract his attention. *"I sent a creative image like the stock image or an image with a valley or clouds, and the hidden message like, 'Hi [Name]' because that's his name. When I sent it to him, saying that you have to squint your eyes to see the message." (P05; CAI, MJ, and BI)* In some cases, recipients expressed interest in further discussions about the generated images, AIGs, or AI in general. These discussions often arose naturally as a result of sharing the images. *"Sharing those images opens up these discussions you have with other people about AI and its capabilities." (P10; CR, CAI, and DI)* Specifically, P15 noted that sharing the image prompted her communication partner to try the AIG and discuss prompting strategies for achieving desired outcomes. *"I shared the Adobe one with my friends and my sister. [...] My sister wanted to create it in a certain way, but this AI kept generating in a different way. She asked me what she should add to the prompt to generate something that she expected. [...] We had that kind of discussion." (P15; AF and BI)*

4.4 Adoption and Perceptions of AI Image Generators

To gain an overview understanding of factors influencing amateur end users' adoption and perceptions of these AIGs (RQ4), we primarily focused on identifying shared perceptions across different tools. These factors were categorized into two main themes: perceived benefits and challenges. We also extracted some distinct considerations that affect tool selection in the creation process.

4.4.1 Perceived Benefits. All participants mentioned the perceived benefits of using AIGs, including 1) breaking barriers to ideation and visualization, 2) expanding creative horizons through unexpected outcomes, 3) optimizing the creative process for efficiency and ease, and 4) achieving original, customized, and varied visual outputs. The following factors motivated users to adopt or continue using the generators:

Breaking barriers to ideation and visualization. Eleven participants found AIGs beneficial for ideation, particularly when lacking a predetermined and fully developed idea or struggling with idea integration in the early stages of the creation process. Participants described these tools as facilitators, aiding in visualizing ideas and transforming them into tangible concepts. *"I like to see AI as a tool that can help me visualize things better because I'm not that great at putting all my ideas. I have many different ideas, and it's so hard to put all those ideas down on one paper." (P07; RW, DE2, and MJ)* Participants also noted that AIGs can help overcome limitations in individual design or artistic skills by generating relatively visually appealing images. These AIGs provide greater accessibility and give amateurs more authority in creation, reducing the sole reliance on professional artists or designers. *"It does a good job of lowering the skill ceiling and barrier to the entry of creating images. [...] Without AI generative tools, I'd either have to find someone that can draw things well, or I'd have to pay an artist." (P11; DE2)* Even participants with more design experiences found the generated visuals to be superior to their own final work.

Expanding creative horizons through unexpected outcomes. Twelve participants valued AIGs for offering a variety of image options that incorporated unexpected visuals, such as unique subject combinations, unanticipated detailed, or uncanny elements, that surpassed participants' own imaginations. For instance, P13 expressed a higher willingness to embrace unexpected ideas and adapt her vision when creating images with AIGs. *"I chose something better because I felt like it looked so much better than what I even imagined it to be. It surpasses that boundary that my imagination puts on me." (P13; MJ, CAI, DE2, etc.)* While the likelihood of achieving desired unexpected outcomes varies from prompt to platform, AIGs occasionally produce surprising and innovative results that foster users' enjoyment in the process. Even users with clear vision benefit from these unexpected outcomes that expand their creativity. *"Because it (image creation process) almost feels like gambling. It's a game that tries to say what you want and it has unexpectedness. [...] Sometimes, you feel like it's rolling a dice, and you get a six. I like those kinds of surprises." (P01; MJ, DE2, and SD)* Notably, most of these participants reported positive experiences with unexpected outcomes stemming from using AIGs like Midjourney or Dall-E due to their generation capability. *"If you're using DALL-E, you kind of have the impression that you'll be pleasantly surprised no matter what you're getting." (P25; DE3 and DE2)*

Optimizing the creative process for efficiency and ease. Nineteen participants emphasized that AIGs significantly optimized the entire creative process by being user-friendly, increasing time efficiency, and reducing personal effort. P12 reflected: *"it's never been easier at any point in my life just to make images of whatever I want." (P12; DE2 and FO)* Specifically, several participants mentioned that achieving the same quality as AI-generated images using traditional methods could be time-consuming, potentially taking hours, weeks, or even months. *"If I did draw this myself, it would probably take me like months, and it would not nearly be as high quality. But with AI, I can just generate multiple images that are fully finished and colored really quickly." (P22; GC)* Additionally, AI-generated images sometimes serve as visuals for post-creation

usage (e.g., sharing with others). AIG shortened the time needed for major editing and revisions.

Achieving original, customized, and varied visual outputs. Most participants initially relied on search engines to find images that aligned with their needs or imagination. However, sixteen participants noted that they occasionally sought more specific visuals that challenged traditional visual representations, such as blending virtual and real elements or depicting humans or objects in unusual layouts. AIGs empowered users to achieve these original, customized outcomes that would have been difficult to find or that might not exist on search or visual discovery engines. *"When I searched on Google, it was really hard to find one that I kind of wanted to use. That's where I think maybe I can use generative AI." (P17; DE2, CAI, and MJ)* Some participants also perceived AI-generated images as original visuals and a way to avoid copyright issues. *"You don't have to worry about like stealing anyone else's work, because it is its own piece." (P25; DE3 and DE2)* On the other hand, many participants highlighted the ability of AIGs to produce abundant and varied range of images that closely align with the prompts as an additional benefit. Particularly, P13 voiced this advantage, which motivated her to continually incorporate AIGs into image creation. *"AI really helps in getting out the plethora of ideas pertaining to the prompt I gave." (P13; MJ, CAI, DE2, etc.)*

4.4.2 Perceived Challenges. All the participants identified several challenges associated with using AIGs, incorporating 1) challenges in developing prompting vocabulary for articulating vision, 2) challenges in communication with AI to make effective modifications, 3) the difficulty of controlling specific image details, 4) inherent limitations of generative models lead to ineffective creation. These perceived challenges may lead to infrequent usage, cross-generator comparison, or a preference for using familiar visual design tools.

Challenges in developing prompting vocabulary for articulating vision. Prompting is the primary method for initiating text-to-image generation in interactions with AIGs. However, nine participants struggled with bringing up comprehensive vocabulary, keywords, or descriptive descriptions to articulate their vision. On one hand, some participants had difficulties putting their vision into words, a challenge that was particularly severe for those whose first language was not English. Observing others' generated outcomes and descriptive prompts could potentially affect participants to reflect on how to improve their own prompting. For example, P06 shared her learning experience in Midjourney's #newbies channel. *"I knew I wanted to generate a Samoyed, but I didn't know how to decorate it (the prompt) and add more descriptive terms until I found how other people do so." (P06; MJ, DE2 and CAI)* On the other hand, while some participants could describe their ideas in their own words, translating these visions into effective AI prompts still remained challenging. Based on some unsuccessful experiences, P04 perceived learning effective prompting as a prerequisite for using AIGs. *"I kind of have a sense, but I don't know how to do the prompt. It's like learning how to prompt. There's a learning curve. That's one part of the challenge." (P04; SD)*

Challenges in communication with AI to make effective modifications. Fourteen participants recognized a discrepancy between their intended vision and the generated outcomes, leading to a sense of lack of control over the results across different AIGs. *"It (AIG) is tricky. Sometimes, you describe it (the prompt) the best you can, but it (the generated image) still doesn't appear the same way in your head." (P22; GC)* The misalignment prompted participants to take action in further fine-tuning and adjustments. Many expressed frustration during this process and indicated a lack of feedback on how to make modifications that AI could effectively interpret. *"It's (AI's) a black box. I don't know which part they accept. When they (AI) didn't accept it, I didn't know how I could address those parts." (P15; AF and BI)* Particularly, fine-tuning is an iterative process that requires experimentation and adjustment. It often involves the back-and-forth and trial-and-error approach to understand how to make adjustments. *"It's (Modifying prompt) not like a linear process. If I add this thing, it will not change a little bit." (P04; SD)* However, the adjustments do not guarantee better outcomes. At this point, participants often weighed their goals and the associated learning curve to determine whether to use AIGs, switch to different AIGs, or stick to their familiar visual design tools.

The difficulty of controlling specific image details. Ten participants encountered challenges in controlling specific image details within AI-generated images. While most of the AIGs produce fully generated images, they often lack the capability for micro-level control. Participants seeking granular control, like image layout, subject movement and positioning, or perspective, frequently encountered limitations with current AIGs. *"The downside is that you can't really control it (AIG). For example, if you're creating a house, you can't control where individual chairs go." (P20; DE3, MJ, DE2, etc.)* Similarly, P01 desired to tailor control over human subjects within the generated images and encountered failure. Due to these difficulties, participants may choose not to use AIGs for highly detailed and specific images and prefer to use familiar tools for greater control. *"I tried using Keynote for those storyboards so there were specific orders like the human should be placed on the left side. It was really hard for me to write in the text." (P17; DE2, CAI, and MJ)*

Inherent limitations of generative models lead to ineffective creation. Eleven participants acknowledged the limitations of AIGs, such as difficulties in generating realistic human faces or avoiding distortions. P08 also highlighted the lack of built-in mechanisms to prevent these unanticipated consequences. *"If you insert some very weird prompts, sometimes they're very disturbing, like over 10 fingers or dogs molded into humans. For some reason, they (AIGs) popped up, something I wasn't looking for. Their limitation is that they are not like humans. They don't have the intelligence to detect their mistakes." (P08; DE2, CR, and DS)* Additionally, participants recognized the potential for embedded AI biases, including ethnicity, to general identity stereotypes that restrict actual usage. For instance, P05 was aware of the AI biases that led her to be more cautious in crafting prompts. *"If you mention the specific ethnicity, they would draw from harmful stereotypes." (P05; CAI, MJ, and BI)*

5 Discussion

Based on our findings, we discuss two different aspects of AIGs: 1) shaping creation workflows: the present and potential future of AIGs and 2) uncovering growing social implications of AIG usage. We also propose design implications and incorporate limitations and future work for further discussion.

5.1 Shaping Creation Workflows: The Present and Potential Future of AIGs

Given the dynamic evolution of AIGs, our study grounds in a user-centered perspective and focuses on structural and conceptual findings that aim to remain impactful both now and in the future. The following discussion section covers high-level creation workflows and detailed creation processes.

5.1.1 About Workflow. Understanding the use of technology in the workflow is essential. By mapping workflows with the prior visual creation process and considering post-actions, our study investigated the AIGs' role across four identified use cases. In the following, we focus on elaborating functional goals, including strategic and discovery-oriented use cases where users evaluate trade-offs in cross-platform or cross-generator usage.

Our observation on cross-platform usage indicated that participants evaluated AIGs based on their inherent trade-offs. This tendency to assess trade-offs of various tools is also reflected in Sanchez's [70] work, where practitioners often combined non GenAI art tools with AIGs, as well as pointed out in the selection of GenAI tools in general [61]. Interestingly, these core trade-off evaluations often arise from unexpected outcomes, which can be perceived as either benefits or challenges. Unexpected results can expand creativity, echoing previous work [39], or turn down unintended consequences. Specifically, in the strategic-oriented use case, AIGs were beneficial in generating nearly polished outcomes and rapidly optimizing the workflow compared to manual methods. However, the difficulty in controlling specific details to align with their clear vision often led to unsatisfactory experiences. This limitation often results in the generated output serving as a temporary visual or prompts participants to use additional tools, such as familiar visual design tools, to refine the images to reach their end goals. On the other hand, in the discovery-oriented use case, participants were engaged in existing workflows and tended to integrate AIGs to optimize the creative process. We found that AIGs primarily support the early stages of the creation process, where unexpected outcomes expand participants' imaginations. Cross-platform usage was driven more by workflow efficiency than AIG limitations. In this case, participants perceived AIGs as supportive aids, resonating with the value of creativity support tools in HCI [31].

Although cross-generator usage was not the primary consideration in creation workflows, we observed scenarios where multiple AIGs were used to accelerate creation. Half of the participants reported experience using multiple AIGs and leveraged them to compare image quality, artistic style, and alignment with their envisioned outcomes. During this process, participants also considered the individual trade-offs of each AIG. For instance, some are concerned about Midjourney's distinctive default image style while appreciating its overall image quality.

In summary, our findings highlighted participants' tendencies for cross-platform and cross-generator usage, as well as the specific stages where integration was most valuable. Specifically, the early stages of the discovery-oriented use case and the mid-stage of the strategic-oriented use case could provide insights into the AIG and creativity community. However, considering the present and future of creativity support tools, we argue that significant challenges still remain in the use of AIG or T2I. Palani et al. [60] emphasized that "seamless" tool integration is essential in the creative process. Yet, how can this integration be effectively achieved in diverse use cases? While visual design tools like Adobe Photoshop and Canva have incorporated T2I features, the question of what constitutes "seamless" integration for different users and usage scenarios still needs to be addressed. Thus, we propose the following design implication:

- **Design implication 1**: Our findings indicated a shift in the perception of AIGs from supportive aids to potential replacement for traditional visual design tools. In particular, the lack of support or frustration occurred primarily in the strategy-oriented use case. Based on these findings, we propose that existing visual design tools could be improved by better supporting users in creating polished final works. By refining their positioning to target professional or functional creative practices, these tools could integrate T2I functionality as a built-in feature. Crucially, they should also offer granular control over AI-generated and user-generated content, along with the flexibility to switch between different generative models, to ensure compatibility with diverse workflows.

5.1.2 About Creation Process. Image creation process represents a micro-level stage where human-AI interactions occur, and we could benefit from a deeper understanding of decision-making and behavioral patterns. In Section 4.2, we identified a generalized usage pattern and three aspects of decision-making when interacting with AIGs. Our following discussion roots in these aspects, examining findings shared across 9 AIGs as well as describing their differences.

Our analysis of "Vision Development and Prompting" revealed diverse strategies to portray participants' initial prompts. Participants using different AIGs demonstrated similarities in content elements, including subjects, actions, scenes, and artistic details, mainly aligning with findings from previous literature [52, 58, 70]. However, we observed distinct decision-making where participants needed to translate their vision into textual input. The perceived challenge of developing the prompting vocabulary may arise during this stage, possibly influencing differences in whether users provide detailed or simple initial prompts. Our results showed that the potential gap between the initial vision and initial prompt could significantly influence the misalignment with generated outcomes, leading to extra effort in fine-tuning. In the "Outcome Evaluation" aspect, we identified several behavioral patterns indicating that evaluation processes were non-linear, often involving assessments within the same iteration or across multiple iterations. When outcomes did not meet the initial vision after several fine-tuning, participants adjusted their expectations and vision accordingly. These adjustments could be positive or negative, depending on the generated outcomes. However, not all existing AIGs support cross-iteration evaluation (e.g., Craiyon) or provide sufficient history record (e.g., DALL-E-2), which limits opportunities for branching out ideas from each output. Regarding the "Adjustment and Fine-Tuning," participants engaged in trial and error, adapting their strategies based on AIG feedback. Interestingly, participants driven by functional goals in image creation, particularly those using simpler prompts, showed greater adaptability in adjusting their ideas to achieve desired outcomes. Many experimented with different themes and branched out from their initial vision into a tree-like structure,

which could be considered another type of usage pattern. However, some participants experienced frustration, particularly those who felt they lacked sufficient guidance and clear cues on how to make effective changes. This highlights the need for better explanations in human-AI co-creation within a visual context [56].

By viewing the creation process as a whole, our generalized usage pattern spans from the initial vision to the final outcome. Similar key decision-making angles, such as image evaluation and prompt refinement, were also observed in the study of Midjourney's prompt journey [52]. However, focusing on sketching activities enabled us to imply a nuanced misalignment between user intentions and generated outcomes, occurring both in transition from vision to initial prompt and the prompt-to-AI translation. Among these AIGs, only several, such as DreamStudio and Midjourney, support different input modalities to compensate for the limitation of text-only input. While we identified this additional level of misalignment, the mechanism behind the difficulty of translating vision into prompt remains unclear. Previous studies have examined user behavior through idea-centric approaches [80] or creative trajectories [41] on broader scales. These approaches could potentially provide insights into user behaviors to improve translation.

Beyond the generalized insights, we argue that it is also crucial to recognize the distinct differences in AIG affordances, ranging from high-level interaction flows and generative models to micro-level functionalities, interface design, the quantity of generated images capacity. For instance, conversational-based (e.g., Dall-E-3 and Midjourney) or demonstration-based (e.g., Dall-E-2 and Canva AI) interactions may lead to varying user experiences and outcomes. Even so, we emphasize the importance of identifying generalized usage patterns and decision-making with different AIGs rather than solely focusing on individual platform differences. As AIGs continue to advance, it is essential to understand in detail how users actually and contextually use these tools. Based on this, we suggest:

- **Design implication 2**: Our analysis revealed a special behavioral pattern where participants evaluation occurred within the same iteration or across multiple iterations. In particular, some participants adopted a tree-like structure and demonstrated a strong desire for evaluation across iterations. To support these behaviors, we suggest that AIGs incorporate a classified history record feature to accommodate divergent and convergent workflows. This feature would allow users to easily review past iterations and group generated outcomes into different themes, enabling more flexible fine-tuning beyond the most recent outputs.

5.2 Uncovering Growing Social Implications of AIG Usage Beyond Functionality and Recreation

Amateur end users with varying design experience demonstrated different motivations when using AIGs, which in turn shaped their corresponding user perceptions. In terms of general perceptions of AIGs, our study shared similar findings on perceived benefits or challenges with previous studies on different types of user groups [39, 40, 52]. However, we still identified distinct aspects unique to amateur users across different AIGs user experiences. Our participants valued AIGs for their ability to break skill barriers in drafting ideas and producing high-quality images. They also emphasized time savings and the expansion of creative possibilities for both functional and recreational purposes. These factors influenced their unique patterns of tool usage and integration (e.g., cross-platform or cross-generator).

To further delve into user goals, our study identified four distinct use cases, broadly categorized into functional and recreational goals (see Section 4.1). While these categories may not fully capture the complexity of user needs and motivations, they shed light on directions for further discussion. For functional purposes, participants prioritized creating practical final work, often partially or non AI-generated, such as posters, slides, and logos. In contrast, recreational purposes focus on generating fully AI-generated content, examples include memes, emojis, and other creative images. Participants may also derive social value by sharing their AI-generated outcomes after creation (see Section 4.3).

Although previous studies [57, 70] have highlighted the social nature of online AIG communities (e.g., Midjourney on Discord), little has been discussed about the recreational goal in the creation process and the social motivations around AIGs. In our findings, especially in the interest-oriented use case, participants showed a strong social inclination to share AI-generated images in direct messages or private group chats with close ones. This behavior highlighted the intention of sharing to foster connections, with images potentially serving as conversation starters. Images share insights align with previous literature suggesting that images act as narrative hooks, influencing emotions, and enhancing connections [32, 47, 74]. Although some existing messaging platforms offer built-in T2I features, such as AI stickers [1], which share similar usage contexts. Our study did not find participants with these experiences.

Another aspect of social implication stems from the characteristics of the image itself. Previous literature has highlighted the richness and open-ended nature of messages conveyed by images. However, the primary challenge in visual communication remains the difficulty in connecting perception, decoding, identifying the relationships between visual elements, and interpretation [34]. Unlike real images, AI-generated images introduce unique synthetic characteristics. Our findings showed that while recipients engaged in visual communication mediated by AI-generated images, their focus often shifted to interpreting these synthetic elements, reflecting the insights of the inherent limitations of generative models. Nonetheless, these unexpected outcomes, without being inherently good or bad, often served as conversation starters and facilitated further discussion. To sum up, we argue that AI-generated images are primarily novelties and are used in restricted social contexts in the current stage. Uncovering the growing social implications around AIG usage, it is worth exploring the changing patterns and emerging social scenarios.

- **Design implication 3**: AIGs possess the potential to deliver social values beyond their primary function, particularly through image sharing in post-creation actions. This observation drives us to suggest conceptualizing AIGs as social tools. The sociocultural context of AIGs has the potential to facilitate interpersonal interactions, enrich collaborative creativity, and encourage community building.

5.3 Limitations and Future Work

Given the rapid progress of GenAI, we were aware of ongoing technical changes when designing and conducting the study. To ensure the adaptability of any potential changes, our interview protocol focused on general user experiences rather than specific AIG features. However, in image creation task analysis, we recognized that interface affordances, such as conversational or demonstrational-based interactions, could potentially influence user behavior in image creation. Future studies are worth investigating how user perceptions and behaviors change over time, and a flexible research approach may be needed. In addition, in response to rapid changes in GenAI era, while our study was conducted in the second half of 2023, it may not fully reflect the latest AIG usage scenarios. Our study still aims to provide structural and conceptual workflows and patterns that remain applicable to future AIG studies. Furthermore, we intentionally conducted a think-aloud task where participants sketched their initial visions to better understand their decision-making and usage patterns. To mitigate potential biases, we verified whether sketching was a common part of their creative process. Although this approach grounded participants' visions and prompts, it may not fully reflect their day-to-day usage, as the gap between initial vision and prompt can differ. Alternative methods, such as diary studies or ethnographic research, may offer more in-depth insights into everyday practices. Lastly, although our study aimed to target amateur end users with diverse backgrounds, we ended up finding that all participants had student roles, which might potentially shape their perceptions. Future research is worth exploring diverse user groups to capture broader perspectives.

6 Conclusion

The advancement of T2I has made AIGs more accessible to end users without domain expertise, leading to their increasing adoption in creative practices. However, the role of various AIGs in the entire creation workflow remains unclear. While prior HCI research has explored user perceptions and interactions with AIGs, detailed user experiences shared across different AIGs are underexplored. To address this gap, we investigated amateur end users' adoption and perceptions of AIGs, focusing on their role throughout the creation workflow. We conducted in-depth interviews with 26 end users across 14 different AIGs, including a think-aloud task. Four distinct use cases emerged, which varied from user goals and initial vision: strategic, discovery, interest, and explorational-oriented. These use cases highlight the involvement of AIGs at different stages of the creative process. Our findings suggest that when users have a functional goal, they often integrate AIGs with other visual design tools to achieve desired outcomes. In contrast, use cases driven by recreational goals mostly influence the social implications of image sharing. We offer design implications for supporting cross-iteration evaluation and suggesting the social role of AIGs.

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Appendix

A Additional information of the participants

Table A.1: Participants' native languages

ID	Native Language
P01	Mandarin
P02	Mandarin
P03	Korean
P04	Mandarin
P05	English
P06	Mandarin
P07	English, Korean
P08	Italian
P09	Bengali
P10	English
P11	English
P12	English
P13	Bengali
P14	Mandarin
P15	Korean
P16	English
P17	Korean
P18	English
P19	English
P20	Korean
P21	Cantonese
P22	English
P23	English
P24	English
P25	English
P26	English

B An Example of Image Creation Task Analysis

We show an example image creation task in Fig. B.1.

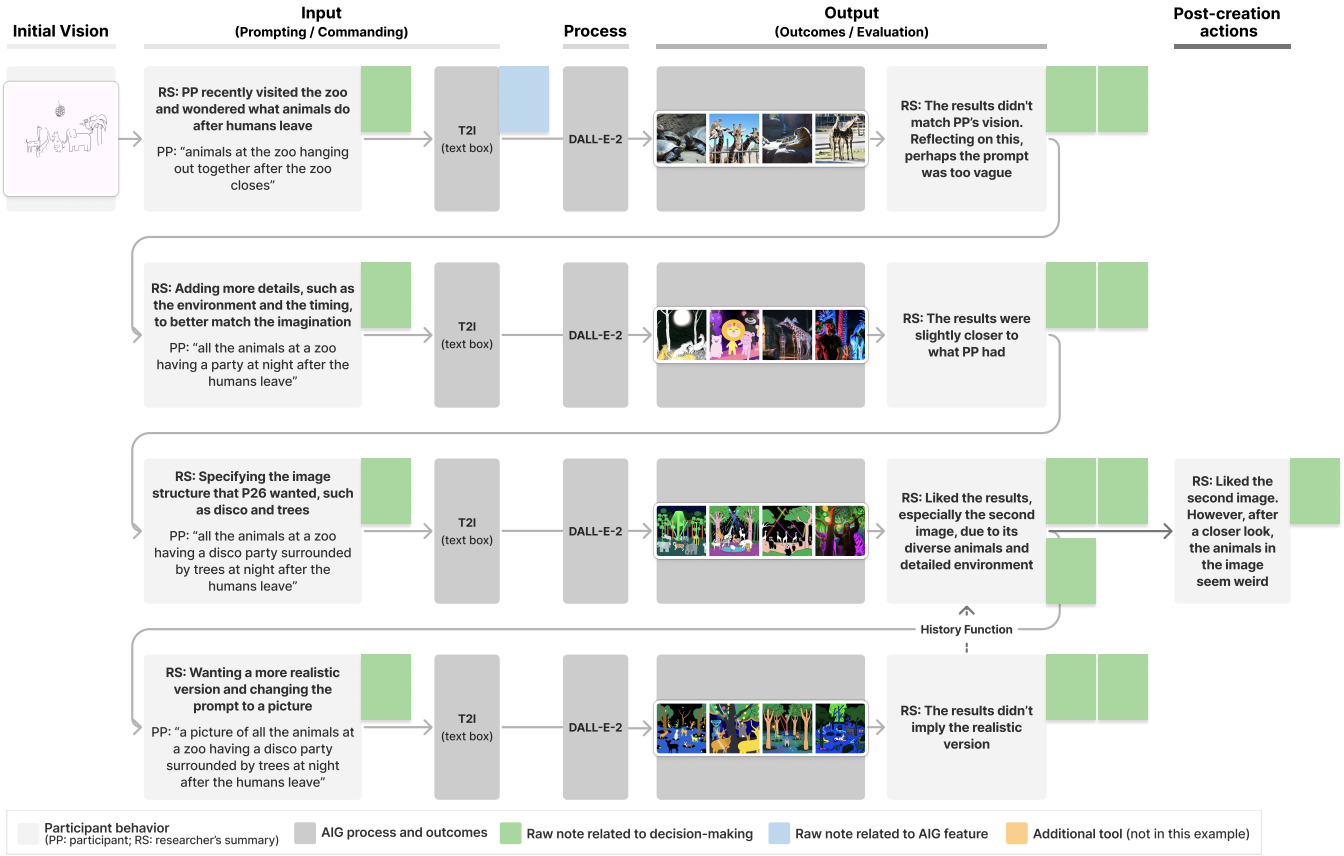

Figure B.1: An example of our image creation task analysis. We examined several elements: 1) sketches, 2) transcripts, 3) chosen image generators, 4) prompts, 5) generated images, and 6) researcher notes and organized them into a diagram for each participant. Each process involves initial vision, input, process, output, and post-creation actions. Different colors in the diagrams represent various data types, and arrows indicate the flow of the creative process. For example, this participant used DALL-E 2, starting with a sketch and verbal description. After typing in prompts, four images were generated per round. Following four rounds of iterations, the participants picked the third round results as the -final work. Best viewed zoomed-in on a display.